

XRPM Direct Verification Repo Blueprint (Locked v1)

1. Monorepo Structure

```
xrpm-login/  
  README.md  
  LICENSE  
  SECURITY.md  
  spec/  
  packages/  
  apps/  
  examples/  
  .github/
```

2. Spec Package

```
spec/  
  XRPM_LOGIN_V1.md  
  proof.schema.json  
  challenge.schema.json  
  canonicalization.md  
  threat-model.md
```

3. Verifier Package

```
packages/verifier/  
  src/  
    index.ts  
    canonical.ts  
    verifySignature.ts  
    deriveAddress.ts  
    errors.ts
```

4. Eligibility Package

```
packages/eligibility/  
  src/  
    index.ts  
    xrplClient.ts  
    checkActivated.ts  
    checkXrpmBalance.ts
```

5. Nonce Store Package

```
packages/nonce-store/  
  src/  
    index.ts  
    memory.ts  
    redis.ts
```

```
postgres.ts
```

6. Web SDK Package

```
packages/sdk-web/  
src/  
  index.ts  
  createChallenge.ts  
  openXrpmApp.ts  
  parseCallback.ts  
  verifyLogin.ts
```

7. UI Components Package

```
packages/ui-react/  
src/  
  LoginButton.tsx  
  QrModal.tsx  
  StatusPanel.tsx
```

8. Partner Demo App

```
apps/partner-demo-next/  
pages/  
  index.tsx  
  api/auth/start.ts  
  api/auth/callback.ts  
lib/  
  nonceStore.ts  
  xrpl.ts
```

9. Optional Verifier Service

```
apps/verifier-service/  
src/server.ts
```

10. XRPM App Requirements

```
apps/xrpm-app/  
docs/  
  signin-deeplink.md  
  proof-generation.md
```

11. Examples

```
examples/  
  express-basic/  
  react-basic/  
  static-html/
```

12. Locked Design Decisions

- Direct verification only (no central auth required)
- Canonical message XRPM_LOGIN_V1
- Proof includes account, pubkey, signature
- Nonce store required (TTL + one-time use)
- XRPM eligibility ≥ 10 enforced